In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
```

In [3]:
```
x=np.array([[-1,1],[-1,-1],[1,-1],[1,1]])
y=np.array([-1,-1,-1,1])
```

In [4]:
```
w=np.zeros(2)
b=0.1
lr=0.1
```

In [5]:
```
def step(x):
    return np.where(x>=0,1,-1)
```

In [7]:
```
for epoch in range(100):
    for i in range (len(x)):
        net=np.dot(x[i],w)+b
        pred=step(net)
        if pred!=y[i]:
            w=w+lr*x[i]*y[i]
            b=b+lr*y[i]
print("Final Weight :- ",w)
print("Final Bias :- ",b)
```

```
Final Weight :-  [0.1 0.1]
Final Bias :-  -0.1
```

In [ ]:
```
pred=
```